Automated Marine Debris Detection using Deep Learning

Project 014: COMP9444 Week 5 Checkpoint (draft)



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Motivation

- Marine debris is a growing threat to ocean ecosystems: it harms marine life and disrupts the natural balance
- Today's surveys and clean-ups rely on people, which is costly, slow and subjective
  - Manual surveys are time-consuming, lack detail and depend on the operator [3]
  - Divers and ROV pilots must find debris and avoid harming flora and fauna [1]
- Vision-equipped robots (ROVs / AUVs) could detect, locate and even remove debris at scale [1]
- Goal: an automated deep-learning detector that finds debris in underwater images



Example frame from the dataset with ground-truth annotation (J-EDI / TrashCan, © JAMSTEC)

Problem Statement

- Task: given an underwater RGB image, detect, classify and localise every debris object
  - Faster R-CNN: class + bounding box
  - Mask R-CNN: class + box + pixel-level segmentation mask
- Classes: debris split by material or object type, plus ROV, plants and animals (so debris must be told apart from marine life)
- Plan: fine-tune COCO-pretrained detectors on TrashCan; compare box vs mask models and Material vs Instance labels
- Evaluate: COCO AP (AP50, AP75, small / medium / large), per-class AP, precision and recall; baseline to beat = Hong et al. [1]
- Why hard: murky water, colour cast, occlusion, small objects, class imbalance

Input
RGB frame



Output
class + box (+ mask)

Literature Review (1/2)

- [1] Hong, Fulton & Sattar (2020). TrashCan: A Semantically-Segmented Dataset Towards Visual Detection of Marine Debris. arXiv:2007.08097
- Problem: no public dataset to train deep nets to detect debris, which is often deformed and non-rigid
- Contribution: 7,212 frames from JAMSTEC J-EDI ROV videos with boxes + instance masks (COCO format); Material and Instance label versions
- Baselines (Detectron2, COCO-pretrained, ResNeXt-101-FPN): Faster R-CNN 10k iterations, Mask R-CNN 15k iterations
- Finding: the Instance version scores higher: visually similar objects share a class (most cans look alike; not all metal objects do)
- Limits: only a baseline; authors expect gains from more data or stronger models

Method	Dataset	AP	AP50	AP75	APS	APM	APL
Mask R-CNN	Instance	30.0	55.3	29.4	23.2	31.7	48.6
Mask R-CNN	Material	28.2	54.0	25.6	24.1	28.7	41.8
Faster R-CNN	**Instance**	**34.5**	**55.4**	**38.1**	**27.6**	**36.2**	**51.4**
Faster R-CNN	Material	29.1	51.2	27.8	28.2	30.2	40.0

Table 1 of [1]. Faster R-CNN on the Instance version is the best baseline (AP 34.5).

Literature Review (2/2)

[2] Valdenegro-Toro (2016). Submerged Marine Debris Detection with Autonomous Underwater Vehicles. RAHA 2016, pp. 1-7

- Early deep-learning approach: a CNN on forward-looking sonar (FLS) imagery to detect underwater debris from an AUV [1]
- Sonar works when light is poor, but it gives far less visual detail than a camera frame (our takeaway)

[3] Ge et al. (2016). Semi-automatic recognition of marine debris on beaches. Scientific Reports 6:25759

- Full-waveform LIDAR scan of a beach; echo amplitude, width and order classified with an SVM into plastic, paper, cloth, metal
- 72 of 87 placed objects detected (82.8%); mean accuracy 75.4% in field validation; glass not detected
- Limits: fixed tripod, beach only, 4 material classes, hand-crafted features

What this means for our project

- Earlier work used sonar or LIDAR plus hand-crafted features or small CNNs; [1] provides the camera-image benchmark (boxes + masks) we build on
- Our focus: end-to-end deep detectors on underwater camera images, comparing box (Faster R-CNN) vs mask (Mask R-CNN) and Material vs Instance labels against the AP 28.2-34.5 baseline

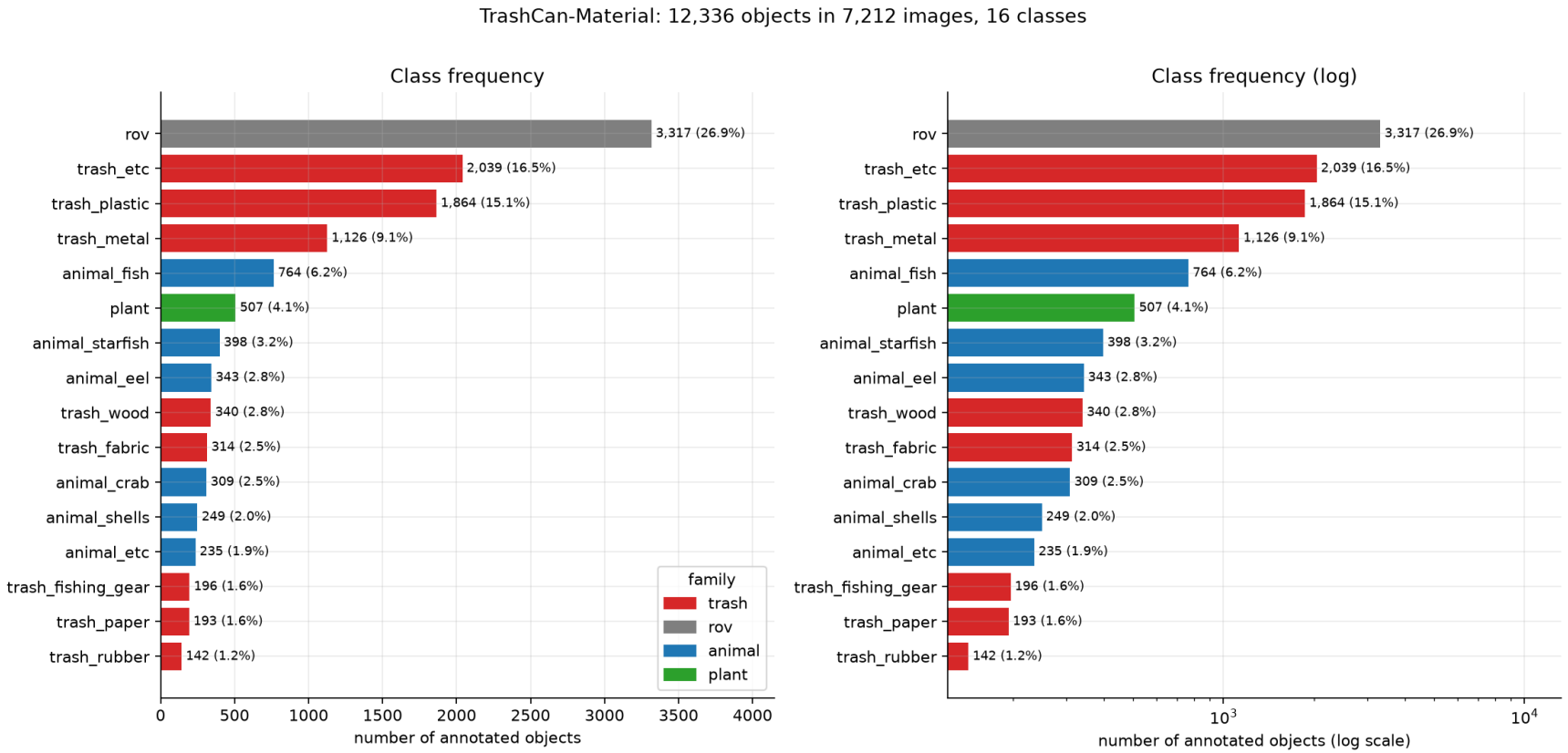
Dataset(s): J-EDI / TrashCan 1.0

- Source: J-EDI (JAMSTEC E-Library of Deep-sea Images), ROV videos since 1982, mostly the Sea of Japan; frames extracted at 1 per second
- Size: 7,212 annotated images (21 annotators, about 1,500 hours, Supervisely)
- Format: COCO JSON: bounding boxes + polygon instance masks (train / val files per version)
- Top-level classes: trash, rov, bio (plant / animal), unknown

Version	Trash classes	Other classes	Rare-class rule
Material	trash_[material], e.g. trash_plastic, trash_metal	rov, plant, animal_[type]	< 50 objects -> trash_etc
Instance	trash_[object], e.g. trash_cup, trash_bag	rov, plant, animal_[type]	< 50 objects -> trash_unknown_instance

Our use: boxes train Faster R-CNN; masks train Mask R-CNN; the two label versions test how label granularity affects AP
Access: University of Minnesota Conservancy (doi:10.13020/g1gx-y834); original videos: JAMSTEC J-EDI deep-sea debris database
Next slides: class statistics computed on the 16-class Material version (12,336 boxes). Instance version still to be analysed.

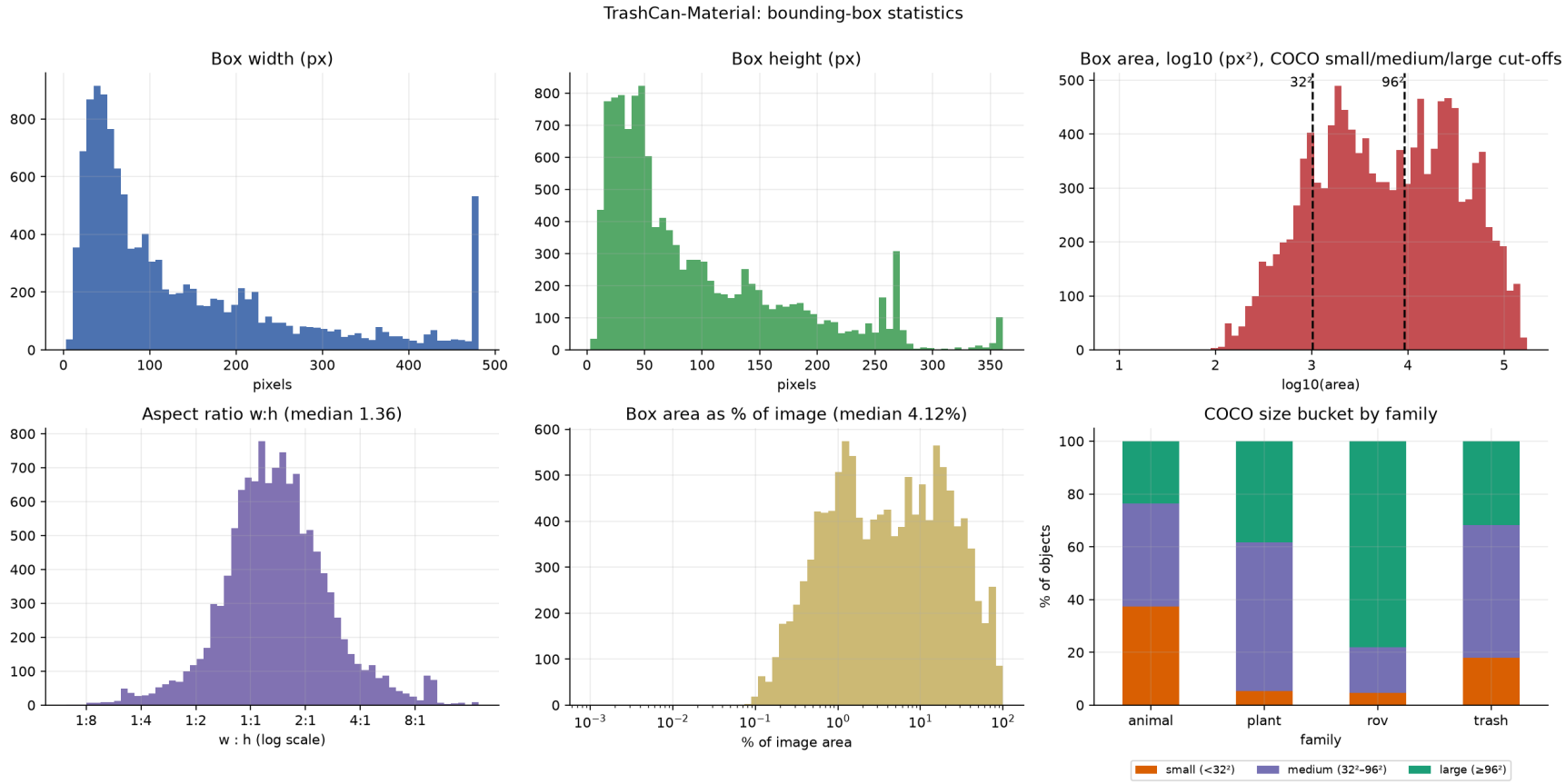
Data Analysis (1/3): class imbalance



- 16 classes, 12,336 boxes, 7,212 images
- Strong imbalance: rov 3,317 (27%) vs trash_rubber 142 = 23x
- Top 3 classes hold 59% of all objects
- Trash classes: 50% of objects; 72% of images contain trash
- trash_etc is the catch-all, so it mixes many object types
- Need class weights / re-sampling and per-class AP

EDA source: community bounding-box export of TrashCan-Material (16 classes). To be re-run on the official COCO files (Material + Instance, with masks).

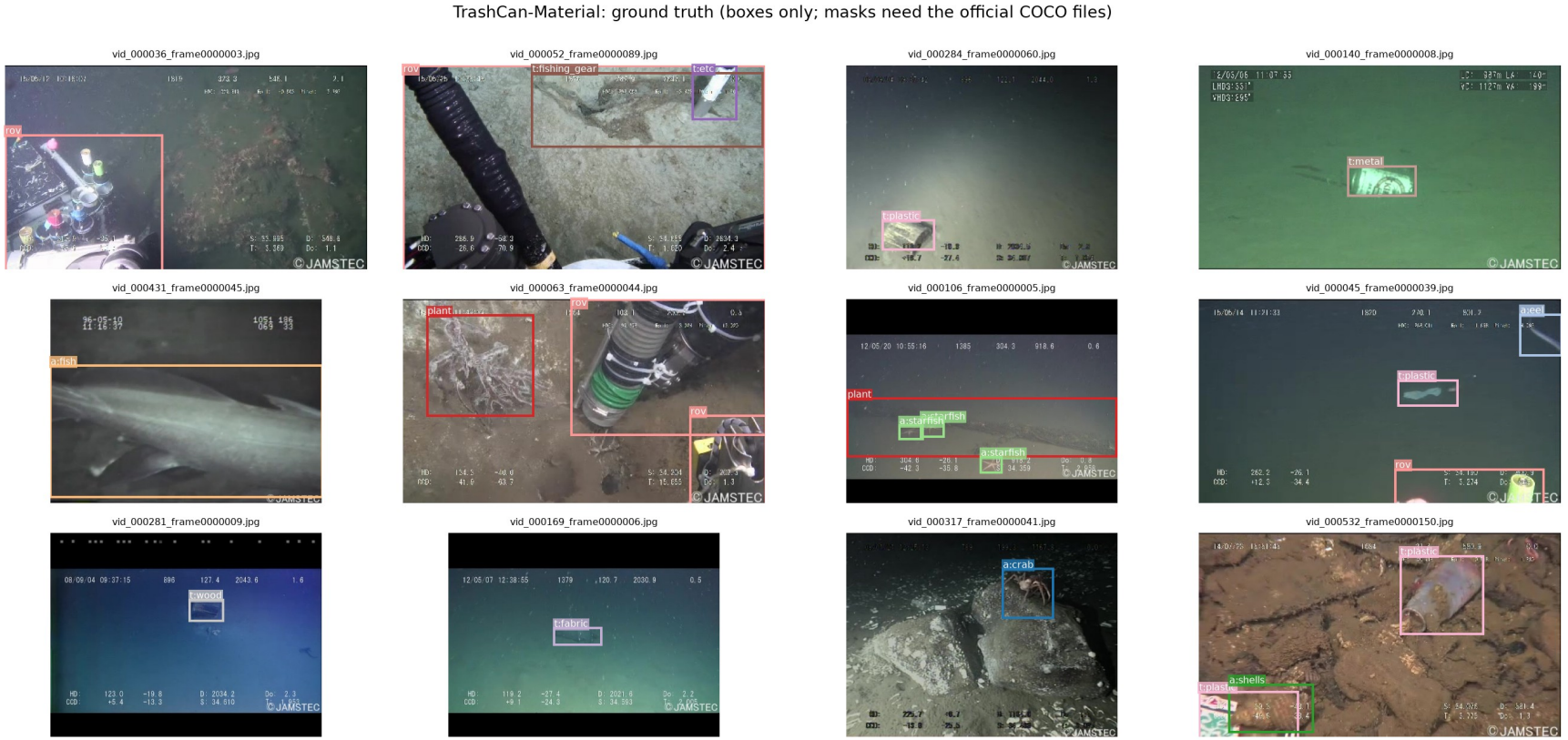
Data Analysis (2/3): objects and box sizes



- Sparse scenes: mean 1.7 objects/image, 58% have one object, max 19
- Trash size: 18% small, 50% medium, 32% large (COCO buckets)
- Median box covers 4.1% of the image; width spans from tens of pixels to the full 480 px frame
- 32% of boxes are elongated (w:h beyond 2:1 or 1:2): choose anchor ratios to match
- Animals are the smallest family; ROV parts are mostly large

EDA source: community bounding-box export of TrashCan-Material (16 classes). To be re-run on the official COCO files (Material + Instance, with masks).

Data Analysis (3/3): visual challenges



- Colour cast: 42% of images have a weak red channel (median red ratio 0.82)
- Murky / dark: 23% low contrast, 15% dark
- Mixed formats: 480x270 and 480x360 px, some with black bars; burned-in telemetry text and a © JAMSTEC mark
- Correlated frames: 312 source videos, median 17 frames each; splits share 0 videos, so no leakage
- Plan: colour / contrast augmentation, video-level CV

Ground-truth boxes on sample frames. Masks come with the official COCO files and are not in this export.

EDA source: community bounding-box export of TrashCan-Material (16 classes). To be re-run on the official COCO files (Material + Instance, with masks).

Method(s)

- Details here …

Results

- Details here …

Discussion

- Details here …

Conclusion

- Write here



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